

Speed Estimation of an Induction Motor Using Artificial Intelligence Algorithms

Degree: Bachelor of Engineering in Electronic Systems Engineering (2025)

Institution: Higher Institute for Applied Sciences and Technology (HIAST), Damascus, Syria

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Evaluation Score: 84%

Dear Professor,

This summary outlines the methodology and outcome of my engineering thesis at HIAST, where I designed and implemented a sensorless speed estimation system for a three-phase induction motor using Artificial Neural Networks (ANN) and validated its performance against an Extended Kalman Filter (EKF).

Stage 1: Dynamic Modeling & Control Simulation

- I derived the non-linear mathematical model of the squirrel-cage induction motor in the synchronous ($d - q$) rotating reference frame using Clarke and Park transformations.
- I designed and simulated a closed-loop scalar V/f speed control system in MATLAB/Simulink, driven by a 3-phase Sine Pulse Width Modulation (SPWM) inverter, providing a baseline to study dynamic behaviors under varied loads and frequency ramps.

Stage 2: AI Neural Network & EKF Design

- I structured a Multi-Layer Perceptron (MLP) network (4-10-5-1 architecture) to estimate motor speed directly from electrical variables.
- To maximize estimation accuracy, I constructed a feature matrix combining ($\alpha - \beta$) active and reactive powers (P, Q), stator voltage magnitudes, and phase currents :

$M_{4 \times n} = [\|V_{\alpha\beta}\| \quad \|I_{\alpha\beta}\| \quad P \quad Q]^T$ comprising voltage/current vector magnitudes and instantaneous active/reactive powers in the ($\alpha - \beta$) frame to reduce neural computational complexity while capturing frequency variations.

- I trained the model using Bayesian Regularization Backpropagation (Levenberg-Marquardt optimization). In simulation, the ANN estimator achieved high accuracy (3 RPM error), significantly outperforming the EKF observer (20 RPM error).

Stage 3: Hardware Testbench & Signal Conditioning

- To validate the approach experimentally, I built an end-to-end data acquisition platform around a 3-phase induction motor testbench.
- I integrated ZMPT101B voltage transformers and ACS712 Hall-effect current sensors.
- To remove high-frequency switching noise from the inverter, I designed custom operational amplifier active filters (OP07) and RC low-pass circuits, scaling and shifting analog signals to fit the (0 – 3.3 V) ADC range of an STM32 microcontroller.

Stage 4: Embedded Firmware & Data Acquisition

- I programmed the STM32 microcontroller in embedded C to sample 5 analog channels simultaneously (2 phase voltages, 2 phase currents, and tachometer feedback for ground-truth comparison).
- Sampling was precisely timed using hardware Timer interrupts, and data was streamed in real-time to MATLAB via USB CDC.

Stage 5: Experimental Testing & Robustness Analysis

- I collected over 72,000 real-world data points under various operating dynamic scenarios, including frequency acceleration/deceleration ramps, step changes, and external mechanical loading.
- Under experimental conditions with electrical and magnetic noise, the trained ANN observer maintained high stability, achieving a speed estimation error within (15 *RPM*).

This project provided me with end-to-end engineering experience combining electric machine theory, signal conditioning circuit design, microcontroller firmware development, and neural network optimization.

Sincerely,

Ali